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International Mathematics Assessments for Schools

2024 / 2025 MIDDLE PRIMARY DIVISION FIRST ROUND PAPER

Time allowed：75 minutes

When your teacher gives the signal, begin working on the problems.

INSTRUCTION AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. No calculators, slide rules, log tables, math stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 20 multiple-choice questions, each with 5 choices. Choose the most reasonable answer. The last 5 questions require whole number answers between 000 and 999 inclusive. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a mathematics assessment, not a test; do not expect to answer all questions.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are filled in. It is your responsibility that the Answer Sheet is correctly coded.

THE ANSWER SHEET

1. Use only pencils.
2. Record your answers on the reverse side of the Answer Sheet (not on the question paper) by FULLY filling in the circles which correspond to your choices.
3. Your Answer Sheet will be read by a machine. The machine will see all markings even if they are in the wrong places. So please be careful not to doodle or write anything extra on the Answer Sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

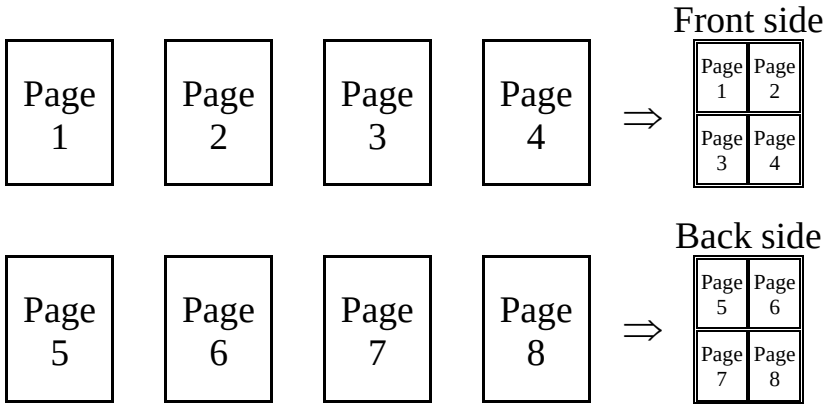
INTEGRITY OF THE COMPETITION

The IMAS reserves the right to re-examine students before deciding whether to grant official status to their scores.

2024 / 2025 MIDDLE PRIMARY DIVISION FIRST ROUND PAPER

Questions 1-10, 3 marks each

1. Andy wants to save paper in printing documents. He does this by printing 8 pages into one sheet of paper as shown in the following configuration below:

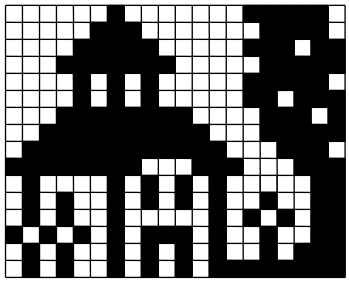


He needs to print one document file, which contains 21 pages, to hand out to 40 persons so that each person gets a complete hard copy of the document. At least, how many sheets of paper will he need to accomplish this task?

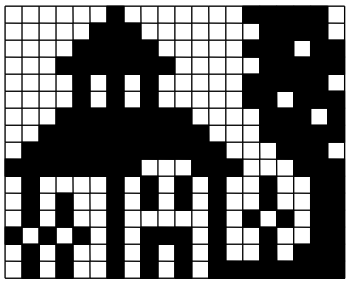
（A）40　　（B）80　　（C）120　　（D）200　　（E）240

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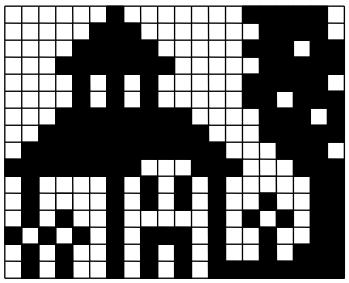
2. Which diagram from options (A) - (E) is EXACTLY the same as the following diagram?



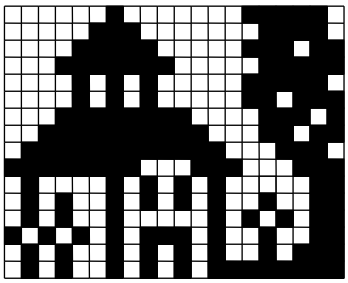
（A）



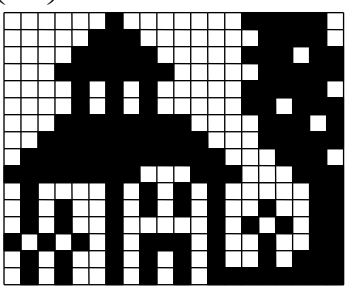
（B）



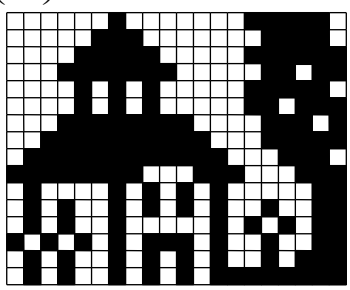
（C）



（D）



（E）



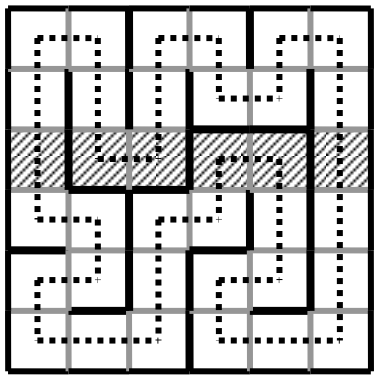
3. Amy is three times as old as Betty this year. Two years from now, Amy will be twice as old as Betty. How old is Amy this year?

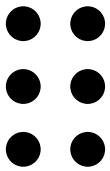
（A）2　　（B）4　　（C）6　　（D）8　　（E）10

4. Twenty-four chairs are equally spaced around a circular table. These chairs are numbered from 1 to 24, in order. Which chair is at the opposite end of the diameter of this round table to chair number 20?

（A）7　　（B）8　　（C）9　　（D）10　　（E）11

5. Six marbles are to be placed on the following maze path shown below so that, when you start from a cell that contains a marble and then travel along the maze paths, you will get to each of the next marbles after passing through the same number of empty cells.

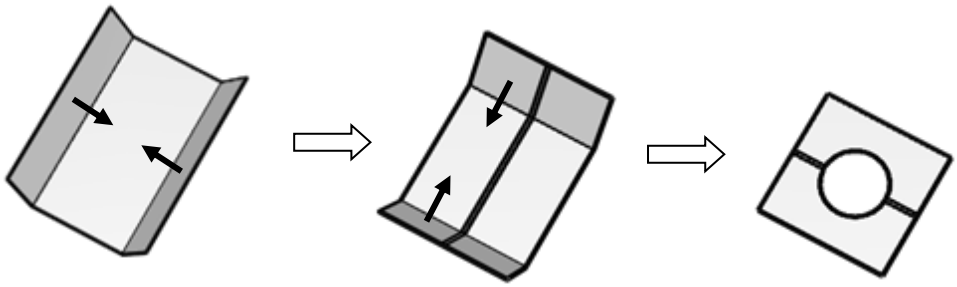




What is the maximum possible number of marbles that can be placed in the shaded row?

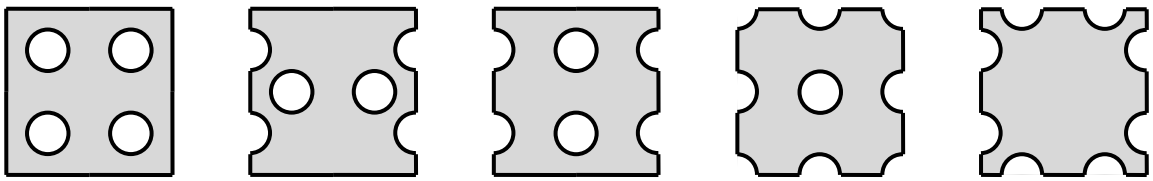
（A）0　　（B）1　　（C）2　　（D）3　　（E）4

6. A sheet of square paper was folded as shown by the arrows, then a hole was punched out in the middle.



Which diagram from options (A) - (E) matches the unfolded square sheet?

（A）　　（B）　　（C）　　（D）　　（E）

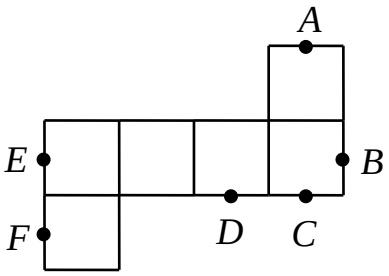


7. Four shuttle buses leave a bus stop all at the same time at 9:00 AM. The first bus returns to this stop every 8 minutes, the second bus every 9 minutes, the third bus every 12 minutes and the fourth bus every 16 minutes. When will be the next time all four buses will be at the bus stop?

（A）10:12 AM　　（B）10:30 AM　　（C）10:36 AM
（D）11:24 AM　　（E）01:48 PM

---

8. In the diagram below, a net of a cube is shown where points *A*, *B*, *C*, *D*, *E* and *F* are midpoints of the edges of the cube. When this cube is constructed, which point does point *F* coincide with?



（A）*A*　　（B）*B*　　（C）*C*　　（D）*D*　　（E）*E*

---

9. Alex has some pencils. He gives half of his pencils to his brother, and then half of what was left to his sister, leaving himself with just 5 pencils. How many pencils did he have at the beginning?

（A）20　　（B）15　　（C）10　　（D）8　　（E）5

---

10. There is a long highway, from county *A* to county *H*, which passes through county *B*, *C*, *D*, *E*, *F* and *G*, in this order. The length of this highway in each county is shown in the table below.

County	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*
Length (in km)	250	600	600	300	500	450	750	700

Which county is the midpoint of this highway located in?

（A）*C*　　（B）*D*　　（C）*E*　　（D）*F*　　（E）*G*

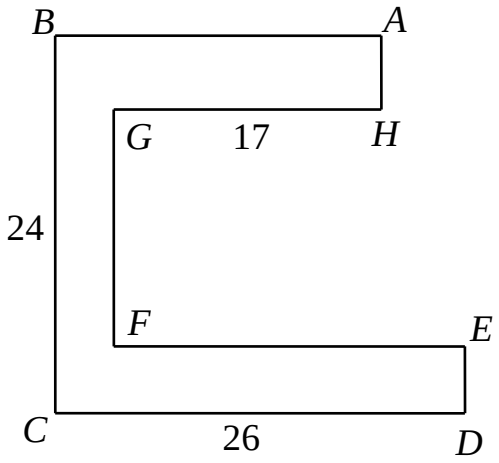
Questions 11-20, 4 marks each

11. How many different 4-digit numbers can be formed by using digits 2, 0, 2 and 5?

（A）8　　（B）9　　（C）10　　（D）12　　（E）24

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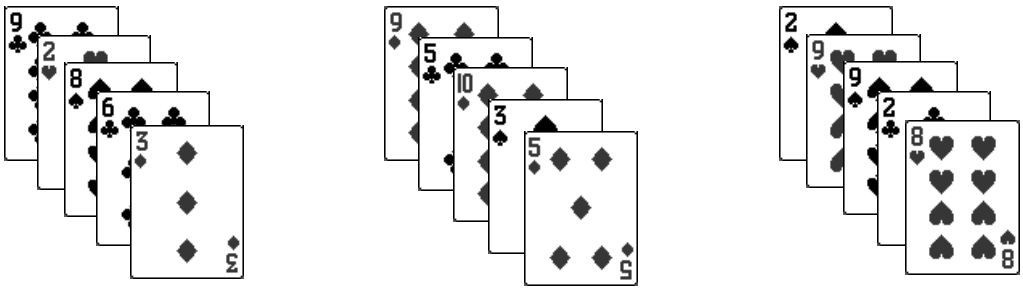
12. In the diagram below, there is an octagon *ABCDEFGH*, where every pair of adjacent edges meet at a right angle. It is known that $BC = 24$ cm, $CD = 26$ cm and $GH = 17$ cm. What is the perimeter, in cm, of octagon *ABCDEFGH*?
Note: The diagram is NOT drawn to scale.



（A）128　（B）130　（C）132　（D）134　（E）144

13. In a strange country calendar, one year consists of 600 days and one week consists of 7 days. The days of the week have the same names and order as our normal calendar. If, in this country's current year calendar, the first day of this year was Friday, on what day of the week would the first day of the next year be?
（A）Saturday　（B）Sunday　（C）Monday
（D）Tuesday　（E）Wednesday

14. Move exactly one card from each pile of five cards to another pile so that the number of cards in each pile is still five and the sum of the numbers in each pile are all equal.



Pile A　Pile B　Pile C

Which card should be removed from pile A?

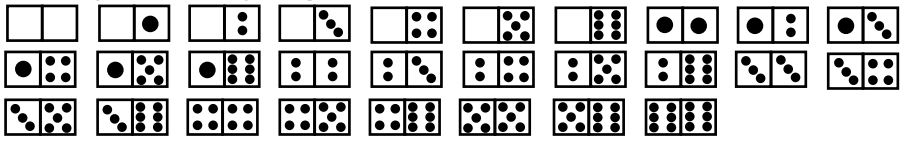
（A）　（B）　（C）　（D）　（E）



15. Given that $a\nabla b = ab + a + b + 3$ and $a\Delta b = ab - a - b + 1$, what is the value of $(2\nabla 5 - 7\Delta 3)\Delta 4$?

（A）17　　（B）18　　（C）19　　（D）20　　（E）21

16. If we connect two unit squares side by side, and mark several dots in either square, we get a domino. If we can only have 0, 1, 2, 3, 4, 5 or 6 dots in each square, there are just twenty-eight different dominoes.



It is known that there are two ways to place a complete domino set of 28 dominoes in the below diagram. However, the sides of the dominoes have been removed and the spots have been replaced by numbers.

4	4	3	3	2	2	1	1
4	4	3	3	2	2	1	1
	5	5	0	0	4	4	
	5	5	0	0	4	4	
	6	6	0	0	6	6	
	6	6	0	0	6	6	
3	3	1	1	5	5	2	2
3	3	1	1	5	5	2	2

Which option among (A) - (E) is a possible correct position of  ?

（A）

4	4	3	3	2	2	1	1
4	4	3	3	2	2	1	1
	5	5	0	0	4	4	
	5	5	0	0	4	4	
	6	6	0	0	6	6	
	6	6	0	0	6	6	
3	3	1	1	5	5	2	2
3	3	1	1	5	5	2	2

（B）

4	4	3	3	2	2	1	1
4	4	3	3	2	2	1	1
	5	5	0	0	4	4	
	5	5	0	0	4	4	
	6	6	0	0	6	6	
	6	6	0	0	6	6	
3	3	1	1	5	5	2	2
3	3	1	1	5	5	2	2

（C）

4	4	3	3	2	2	1	1
4	4	3	3	2	2	1	1
	5	5	0	0	4	4	
	5	5	0	0	4	4	
	6	6	0	0	6	6	
	6	6	0	0	6	6	
3	3	1	1	5	5	2	2
3	3	1	1	5	5	2	2

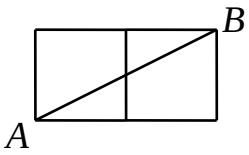
（D）

4	4	3	3	2	2	1	1
4	4	3	3	2	2	1	1
	5	5	0	0	4	4	
	5	5	0	0	4	4	
	6	6	0	0	6	6	
	6	6	0	0	6	6	
3	3	1	1	5	5	2	2
3	3	1	1	5	5	2	2

（E）

4	4	3	3	2	2	1	1
4	4	3	3	2	2	1	1
	5	5	0	0	4	4	
	5	5	0	0	4	4	
	6	6	0	0	6	6	
	6	6	0	0	6	6	
3	3	1	1	5	5	2	2
3	3	1	1	5	5	2	2

17. Two identical small squares are placed side by side to form a bigger rectangle without overlapping, as shown in the diagram below.



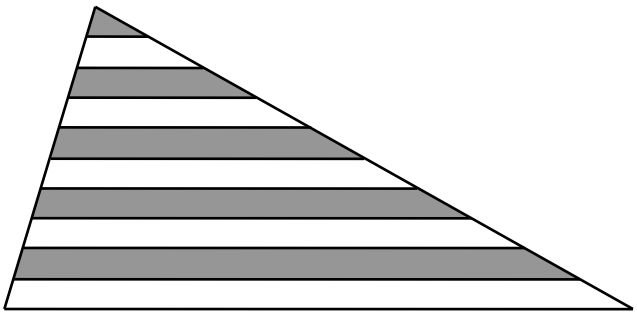
If AB is a diagonal of the rectangle and $AB = 10$ cm, what is the area, in cm^2, of one small square?

（A）10　（B）20　（C）30　（D）40　（E）50

18. In a student camp there are 280 students, where 40% are boys. A school bus brought more students, which contained an equal number of boys and girls, to the camp. If the percentage of boys in the camp is now increased to 42%, what is the total number of students that were on the bus?

（A）35　（B）49　（C）56　（D）70　（E）112

19. In a triangle, we draw some segments that are parallel to the base, which divide the other sides into 10 equal parts, as shown in the diagram below. If the area of this triangle is 1000 cm^2, what is the total area, in cm^2, of the shaded regions?



（A）400　（B）425　（C）430　（D）450　（E）500

20. By writing each of the positive integers starting from 1 from left to right, we get the following sequence:

12345678910111213 14...

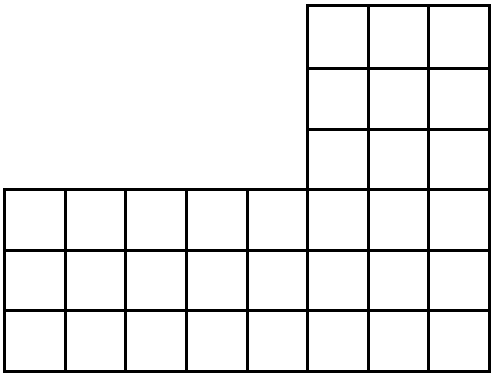
In the sequence, the 15th digit is 2. What is the 2025th digit in this sequence?

（A）1　（B）2　（C）5　（D）7　（E）9

Questions 21-25, 6 marks each

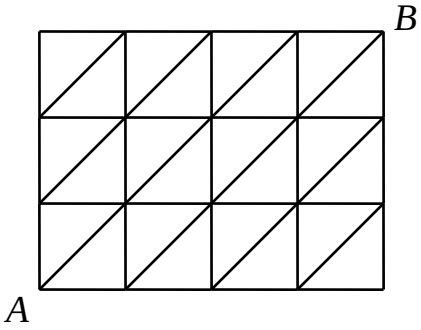
21. Let a, b and c be three different digits so that $\overline{aac} + \overline{cca} = \overline{bbb}$. What is the second largest possible value of $\overline{abc}$?

22. How many squares can be formed from the grid lines in the following diagram given that all adjacent grid lines are equally spaced?

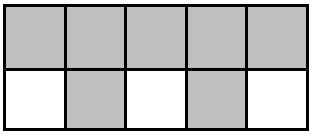


23. There are three positive integers a, b and c. If the greatest common divisor of a and b is 20, while the greatest common divisor of b and c is 25, what is the sum of the two smallest possible values of the greatest common divisor of a and c?

24. How many paths are there to travel from point A to point B by only moving along the segments of the following grid if you are only allowed to go upwards, rightwards, or diagonally up?



25. The diagram below shows a 2×5 board with 7 unit cells coloured gray. In how many ways can one fill the numbers 1, 2, 3, 4, 5, 6 and 7 into the gray cells, where each cell can be filled with exactly one number, and each number is used exactly once, such that the numbers on the first row are increasing from left to right and the numbers on the second column and on the fourth column are increasing from top to bottom?



* * *